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March 6, 2026

To the Editors-in-Chief *Advanced Engineering Informatics* Elsevier

Subject: Submission of Original Research Article “GeoVault: Leveraging Human Spatial Memory for Secure Cryptographic Key Management”

Dear Editors,

I am pleased to submit the manuscript entitled “**GeoVault: Leveraging Human Spatial Memory for Secure Cryptographic Key Management**” for consideration as an original research article in *Advanced Engineering Informatics*.

The central challenge addressed in this work is the persistent failure of human-centered security systems due to the mismatch between cryptographic requirements and human cognitive constraints. While traditional systems rely on fragile linguistic memory (passwords), our research proposes a novel informatics framework that utilizes **human spatial cognition**—a robust, evolutionarily grounded memory system—as a source for deterministic key derivation.

We believe this paper is an excellent fit for the Aims and Scope of *Advanced Engineering Informatics* for several reasons:

- **Engineering Informatics Applications:** We apply geospatial discretization and discretization algorithms (like What3Words and Google S2) to solve a critical data-security problem in engineering management.
- **Human-Computer Interaction (HCI):** The work explores how the hierarchical structure of human spatial memory can be formalized into an information-theoretic model to improve the usability of secure systems.
- **Knowledge Representation:** We develop a formal entropy model that captures semantic and clustering biases in human location choice, providing a bridge between cognitive psychology and computational security.
- **Computational Hardening:** We utilize memory-hard key derivation (Argon2id) to enforce a hardware-aware asymmetry, protecting user data against modern GPU-based brute-force attacks.

By demonstrating that spatial memory can provide a higher effective entropy floor than traditional alphanumeric strings, this study offers a transdisciplinary solution that aligns with the journal’s focus on advanced computing and human-centered engineering tasks.

This manuscript has not been published elsewhere and is not under consideration by another journal. All authors have approved the final version and agree with its submission. We have no conflicts of interest to disclose.

Thank you for your time and for considering our work.

Sincerely,

Assis. Dr. Marko Corn